

Labwork 2	
Course	<i>Modelação de Dados em Engenharia</i>
	2025/2026
Objectives	Model and implement a logistics network with mobile robots
Classes	5 classes 6 extra hours
Submission	2026/05/23
Objectives: 1. Fact Modelling a. Represent the problem using an appropriate fact structure in the knowledge base. 2. Definition of Inference Rules a. Perform queries over the facts defined in the knowledge base. b. Infer knowledge based on the defined facts and rules. 3. Use of Modelling Tools a. Use SWI-Prolog to develop and test the knowledge base. b. Enable interactivity between the user and the knowledge base using SWI-Prolog. 4. Integration of the Knowledge Base with Other Software a. Collect real-time data for the knowledge base using the MQTT protocol. b. Establish communication between SWI-Prolog and a Java application using the JSON format. c. Enable interactivity between the user and the knowledge base through a Java application. 5. Documentation and Evaluation. a. Complete the worksheet on Moodle, documenting decisions and results.	

Objectives

This assignment aims to provide students with exposure to the concepts of knowledge modelling.

Problem Description

The increasing demand for more efficient, faster, and sustainable distribution systems has driven the development of intelligent solutions in the field of logistics, integrating monitoring technologies, communication networks, and decision-support mechanisms. In this context, the goal is to **model an Autonomous Logistics Network**, composed of **suppliers**, **distribution points** (hubs), **customers**, **charging stations**, and autonomous mobile **robots**, with the aim of optimizing product transportation and delivery processes.

Each **supplier** can provide **different** types of **products**, which are **routed either directly** to customers **or** to **intermediate distribution points** responsible for consolidating and redistributing orders. These points may operate as logistics hubs, enabling more efficient management of delivery flows and better adaptation to demand across different areas.

Autonomous **robots**, including **ground** delivery robots, **drones**, and **autonomous vehicles**, are equipped with monitoring systems that allow **real-time tracking** of their **operational status** (Figure 1). The collected **data includes location, battery level, mission status** (idle, busy, or charging), and **transported load**. This data is **periodically updated** and can be **integrated** into an **intelligent decision-support system**, enabling, for example, the identification of critical situations (such as low battery), task redistribution, or optimization of the use of available resources.



Figure 1 – Illustrative example of monitoring data for autonomous robots.

The logistics **network** also **includes various customers** (representing the final delivery destinations), **distribution points** (which act as intermediate elements in the transportation chain), and **charging stations** (allowing robots to recharge along their routes). These elements

are associated with different levels of demand and utilization, which must be considered in delivery and operations planning, ensuring both service continuity and the system's energy efficiency.

The transportation of products between suppliers, distribution points, and customers is carried out by autonomous robots with different characteristics, namely:

- Load capacity
- Robot type (ground, drone, autonomous vehicle)
- Energy autonomy
- Travel speed
- Estimated energy consumption per route

In this context, the system should incorporate efficiency and sustainability criteria in the process of route selection and robot allocation, prioritizing solutions that reduce energy consumption, delivery time, and the number of required trips. Other relevant variables should also be considered, such as the distance between points, estimated travel time, compatibility between robot type and delivery type, available capacity at any given moment, and the need to pass through charging stations along the route.

The logistics network will be represented by a structure such as the one shown in Figure 2, consisting of interconnected nodes (suppliers, distribution points, customers, and charging stations) and links with attributes such as distance, travel time, type of route (urban, aerial, mixed), and energy cost. Modelling should allow, for example, the calculation of optimized routes between different points in the network, according to criteria such as:

- Shortest distance
- Shortest delivery time
- Lowest energy consumption
- Highest delivery priority based on urgency or importance of the request
- Optimization through intermediate distribution points whenever it helps reduce the number of trips
- Route planning including charging stations, ensuring that robots have sufficient energy to complete their missions

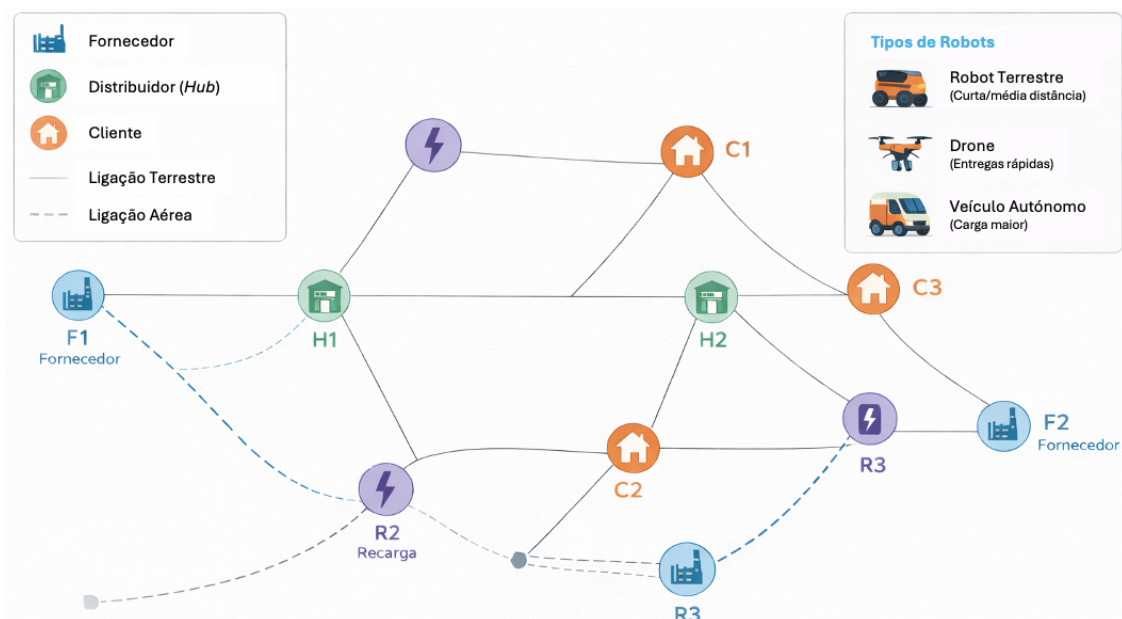


Figure 2 – Illustrative example of an autonomous logistics network.

In this context, the aim is to develop a management system to assist the operator or administrator of the logistics network in querying and configuring different system properties. The system should allow, for example, the execution of customised queries, the visualisation of organised results (in lists or tables), the updating of the knowledge base with new data (new robots, new orders, status changes), and the export of results (such as routes or decisions) in JSON or a similar format.

To achieve this, the system will integrate an inference engine, responsible for determining optimised solutions for delivery planning and for supporting decision-making based on the criteria defined by the operator. The solution will be developed through the modelling of a knowledge base and inference rules, using a logic programming language.

The knowledge base and the corresponding system should provide an interface through which the operator can view different elements of the network, such as: which suppliers and customers exist, which robots are available and their current status, which orders are pending, which robots have critical battery levels, which charging stations are available, or which routes are the most suitable for a particular delivery, thereby enabling efficient and intelligent management of the autonomous logistics network.

The system should be divided into two phases:

Phase 1 – Knowledge Base Management

For knowledge base management, the goal is to create a Prolog program that allows the knowledge base to be edited (inserted, removed, and modified), defining suppliers, distribution points, customers, autonomous robots, charging stations, etc. This phase forms the foundation of the system's knowledge representation, enabling the construction of all the data required to support the inference logic in the following phase. In this first phase, the system should be able to:

✓ Add/remove/edit:

- Suppliers, distribution points (hubs), and customers
- Autonomous robots
- Charging stations and their respective characteristics
- Connections between network nodes
- Delivery orders
- Robot status

✓ Simple queries:

- List available robots in a given area
- List suppliers, distribution points, and customers
- Check the current status of a robot
- List pending delivery orders
- List available charging stations in a given region
- Check connections between network nodes
- View the battery level and location of robots

Phase 2 – Inference

In the second phase of development, the goal is to explore the system's potential as a logical inference engine, capable of answering complex questions about the operation of the logistics network, the status of the robots, and delivery efficiency. Some examples include:

✓ **Queries and inferences about robots and system status:**

- Identify robots with battery levels below a critical threshold (e.g. battery < 20%)
- Determine which robots need to pass through a charging station before completing a delivery
- Present the evolution of a robot's status over time (location, battery, tasks)
- Identify overloaded or underutilised robots

✓ **Simple and cross-multicriteria queries and inferences about routes and distribution:**

- Determine the shortest route between a supplier and a customer
- Determine the route with the lowest estimated delivery time
- Determine the route with the lowest energy consumption, considering the robot type and the route
- Determine the ideal route passing through an intermediate distribution point, if this reduces the number of trips or improves logistics efficiency
- Determine routes that include charging stations, ensuring that the robot can complete the mission without running out of battery
- List customers with the highest number of orders or highest priority
- List the best robot for a given delivery, taking into account simultaneously: load capacity, autonomy, location, robot type, and current status
- Determine which robots can carry out a set of grouped deliveries while minimising the number of trips

Note: These are only some examples of what the system may offer. The student will only need to implement what is required in the following section regarding the functional requirements.

Functional Requirements

The study and implementation of the problem proposed in this assignment will allow the exploration of the main concepts of logic programming, namely: atoms, predicates, facts, rules, inference, knowledge bases, and the Prolog language. One of the central objectives is to understand the difference between conventional approaches, based on imperative languages, and logical paradigms, which are more oriented towards knowledge representation and automated reasoning. In this context, building a logical system will provide an opportunity to apply these concepts to a real and multidimensional problem: the sustainable management of an intelligent agricultural network.

Table 1 Functional requirements for the development of the knowledge base

Functional Requirements	Description
RF 1	Add/modify/remove a network node.
RF 2	Add/modify/remove autonomous robots.
RF 3	Add/modify/remove connections between network nodes.
RF 4	Register and update the status of robots.
RF 5	Create, update, and remove delivery orders.
RF 6	List available robots in a given area or region.
RF 7	Check the current status of a specific robot.
RF 8	List available charging stations in a given area or near a robot.
RF 9	Determine whether a robot can perform a given delivery, considering capacity, autonomy, and current status.

RF 10	Determine the shortest route between two nodes in the network.
RF 11	Determine the route with the lowest energy consumption, considering the robot type and the path.
RF 12	Determine routes that include intermediate distribution points when advantageous.
RF 13	Determine routes that include charging stations, ensuring sufficient autonomy to complete the delivery.
RF 14	Select the most suitable robot for a delivery based on multiple criteria (capacity, autonomy, location, robot type, status).
RF 15	Identify robots capable of executing a set of grouped deliveries. (The system should be able to determine which autonomous robots can execute a set of delivery requests in a single mission, instead of performing separate individual deliveries, i.e., one robot delivering a set of orders.) For this functionality, factors such as load capacity, autonomy, route feasibility (possible paths connecting delivery points), and the robot's current state must be considered.
RF 16	Define and implement an additional functionality proposed by the students.
RF 17	Register and update the state of robots using data received via MQTT.
RF 18	Create a server in SWI-Prolog providing the functionalities above and develop a Java application.

Development

Students should use the SWI-Prolog development environment and VS Code (or another Java IDE of their choice) to develop the proposed solution. The assignment should be structured according to the following stages:

1. Use of Modelling Tools
 - Use SWI-Prolog software to create and test the knowledge base.
 - Allow user interaction to perform queries, updates, and simulations on the knowledge base.
2. Fact Modelling
 - Represent producers, farms, sensors, distributors, transport operators, and network connections.
 - Associate producers and sensors with farms and record periodic readings.
 - Represent product requests made by distributors.
 - Record water consumption by crop or by farm.
3. Definition of Inference Rules
 - Define all inference rules that satisfy the functional requirements in Table 1.
4. Integration with Other Systems
 - Collect sensor data in real time via the MQTT protocol (simulated or real).
Integrate the knowledge base with a Java application, exchanging data via JSON.
 - Provide an interface that allows the visualisation of network elements, sensor readings, and route suggestions.
5. Documentation and Evaluation
 - Submit the project and complete a worksheet on Moodle, justifying decisions and presenting the results.
 - Consult Table 2 to learn about the assessment criteria applied to the assignment.

Reflection Elements

This section consists of more reflective considerations that could serve as “inspiration” for writing a report (however, no report submission is required for this assignment). At the end of the work, the following questions could be addressed:

- What is meant by modelling and the notion of a model.
- The concept of abstraction (illustrated with specific details from the presented problem).
- The usefulness of modelling in the context of Engineering.
- The ambiguous, redundant, and incomplete nature of natural language as a modelling formalism.
- The characteristics that a modelling formalism (or language) should have to adequately represent models.
- Knowledge representation and what is meant by a Knowledge Base.
- The notion of inference/reasoning.

- Logical languages.

Work Plan

1st class:

Introduction to the Prolog language.
Installation of the SWI-Prolog development environment.
Examples of fact and rule representation, queries, and recursion.

2nd class:

Presentation of the problem.
Modelling of the knowledge base for the proposed assignment.

3rd class:

Development and implementation of the functional requirements defined in the practical work.

4th class:

Development and implementation of the functional requirements defined in the practical work.

5th class:

Finalisation of the assignment.
Submission (Moodle).

Submission (Moodle)

Students must submit, via Moodle, a zip/rar file containing the source code (including facts/rules that model the problem), as well as a demonstration video of approximately 5 minutes. If additional components are developed (e.g., an external interface), the corresponding project should also be submitted.

This assignment requires the mandatory completion of a questionnaire on Moodle, the submission of a video (which replaces the in-class presentation and discussion), and a submission point for the complete project. Submissions made via email will not be considered.

Evaluation

It is subject to an oral presentation, in which all group members must be present (absent students will receive 0 in the practical component assessment). Although the project is carried out in groups, the grade for this component will always be individual. The assessment will take into account the project itself, performance in the oral presentation, the report, and an individual evaluation.

Table 2 presents the grid with the assessment criteria for the assignment.

Table 2 – Evaluation

Itens	Descrição	Evaluation
Functional Requirements	RF 1 Add/modify/remove a network node.	0.5
	RF 2 Add/modify/remove autonomous robots.	0.5
	RF 3 Add/modify/remove connections between network nodes.	0.5
	RF 4 Register and update the status of robots.	0.5
	RF 5 Create, update, and remove delivery orders.	0.5
	RF 6 List available robots in a given area or region.	1.0
	RF 7 Check the current status of a specific robot.	0.5
	RF 8 List available charging stations in a given area or near a robot.	0.5
	RF 9 Determine whether a robot can perform a given delivery, considering capacity, autonomy, and current status.	0.5
	RF 10 Determine the shortest route between two nodes in the network.	1.0
	RF 11 Determine the route with the lowest energy consumption, considering the robot type and the path.	1.0
	RF 12 Determine routes that include intermediate distribution points when advantageous.	1.0
	RF 13 Determine routes that include charging stations, ensuring sufficient autonomy to complete the delivery.	1.5
	RF 14 Select the most suitable robot for a delivery based on multiple criteria (capacity, autonomy, location, robot type, status).	1.5
	RF 15 Identify robots capable of executing a set of grouped deliveries. (The system should be able to determine which autonomous robots can execute a set of delivery requests in a single mission, instead of performing separate individual deliveries, i.e., one robot delivering a set of orders.) For this functionality, factors such as load capacity, autonomy, route feasibility (possible paths connecting delivery points), and the robot's current state must be considered.	2.0
	RF 16 Define and implement an additional functionality proposed by the students.	1.5
	RF 17 Register and update the state of robots using data received via MQTT.	1.0
	RF 18 Create a server in SWI-Prolog providing the functionalities above and develop a Java application.	1.5
Video	Visualization of the developed functionalities	1.0
Discussion	Group performance (project, oral discussion performance, report)	2.0
Total		20.0
Failure to submit one of the items		-20.0

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